

Technical Report

1. Assumption

Before the data preprocessing, we have some necessary assumptions below.

1. Since the question does not give the number of participants and their events for each country in 2028, we will rely more heavily on the data for each country in 2016, 2020 and 2024 when predicting the number of gold, silver and bronze medals and total medals won by each country in 2028.
2. Not taking into account the number of medals for East Germany, West Germany, Russia and the USSR.
3. Countries and their athletes that have participated in any of the three previous Olympic Games will still participate in the 2028 Olympic Games.

2. Data Preprocessing

2.1 Research Background and Data Overview

The core dataset of this study contains over 250,000 historical Olympic athlete participation records, with each entry including fields such as athlete name, gender, team affiliation, National Olympic Committee code (NOC), year, host city, sport, event, and medal outcome. The data spans from the first modern Olympics in 1896 to the 2024 Paris Games.

2.2 Data Preprocessing Pipeline

The Olympic medal prediction data preprocessing pipeline in this project consists of the following four key phases.

First, basic data cleaning is performed by loading summer Olympic athlete records (from 2000 onward) to extract valid country/region lists, while simultaneously loading historical medal data. Invalid records are filtered through country code matching to ensure data consistency, with missing values in medal data uniformly filled with zeros. The data is then aggregated by "year-country" dimensions to calculate total gold, silver, and bronze medals.

In the second phase, time-series features are constructed using a sliding window technique that utilizes medal data from the past three Olympics as feature variables (automatically reverting to historical averages when fewer than four data points are available). Model performance is evaluated through leave-one-out validation: XGBoost models are trained on 2000-2016 data to predict 2020 medal counts, with metrics like MAE and RMSE calculated.

The third phase generates 2028 predictions, applying XGBoost time-series forecasting for countries with sufficient historical data (≥ 4 participations) and historical averaging for others. Predictions are disaggregated into gold/silver/bronze based on each country's historical medal ratios.

The entire process ensures reproducibility through fixed random seeding (42). Notably, this implementation focuses on fundamental time-series modeling preprocessing based on historical medal data, excluding more advanced operations mentioned in the original description such as athlete name processing, IQR outlier detection, or socioeconomic indicator integration.

Key adaptations:

1. Removed unimplemented components (PCA dimensionality reduction, external data integration)
2. Highlighted actual feature engineering (3-year sliding window)
3. Emphasized the two prediction logic paths (XGBoost/historical average) and their triggers
4. Maintained core evaluation methodology (leave-one-out validation) and output structure
5. Clearly noted differences from the original description (e.g., simple zero-filling for missing values)

3. Construction of Models

For this project, we mainly used three models: **multivariate linear regression**, **xgboost** and **random forest**. In making the predictions, the model takes into account the number of prizes won by each country's historical participation as well as the performance of athletes in the last three Olympic Games and the num of medal in the history. For linear regression, we can formulate the equation:

$$\text{Total}_i = \beta_0 + \beta_1 \cdot \text{HM}_i + \beta_2 \cdot \text{H}_i + \beta_3 \cdot \text{AP}_i + \beta_4 \cdot \text{EV}_i + \epsilon \quad (1)$$

Variable Definition:

$$\beta_1 \cdot \text{HM}_i = \beta_{11} \cdot \text{Past_Total_Average}_i + \beta_{12} \cdot \text{Past_Total_Sum}_i \quad (2)$$

Past_Total_Sum: The country's level of participation in the competition.

Past_Total_Average: The strength of the country's ability to win medals.

$$P_{i,j} = \frac{M_{i,j}}{A_{i,j}} \quad (3)$$

The average performance over the past three Olympic Games is:

$$\text{AP}_i = \frac{P_{i,j-2} + P_{i,j-1} + P_{i,j}}{3} \quad (4)$$

$$N_i = w_1 \cdot N_{\text{new}} + w_2 \cdot N_{\text{cancel}} \quad (5)$$

N_{new} : The impact score of newly added events.

N_{cancel} : The impact score of canceled events.

w_1, w_2 : Weight parameters for each factor.

Explanation of β Coefficients

β_0 : Intercept, which represents the predicted total medal count ($Total_i$) when all variables ($Past_Total_Avg_i$, $Past_Total_Sum_i$, AP_i) are zero. It provides the baseline level of total medals.

β : Used to indicate the strength of the relationship between the dependent and independent variables and whether or not they are positively correlated

ϵ : Error term, which captures variability in $Total_i$ that is not explained by the model.

By defining the multiple linear regression variables, the fitted results can be obtained and shown in **Figure 1**.

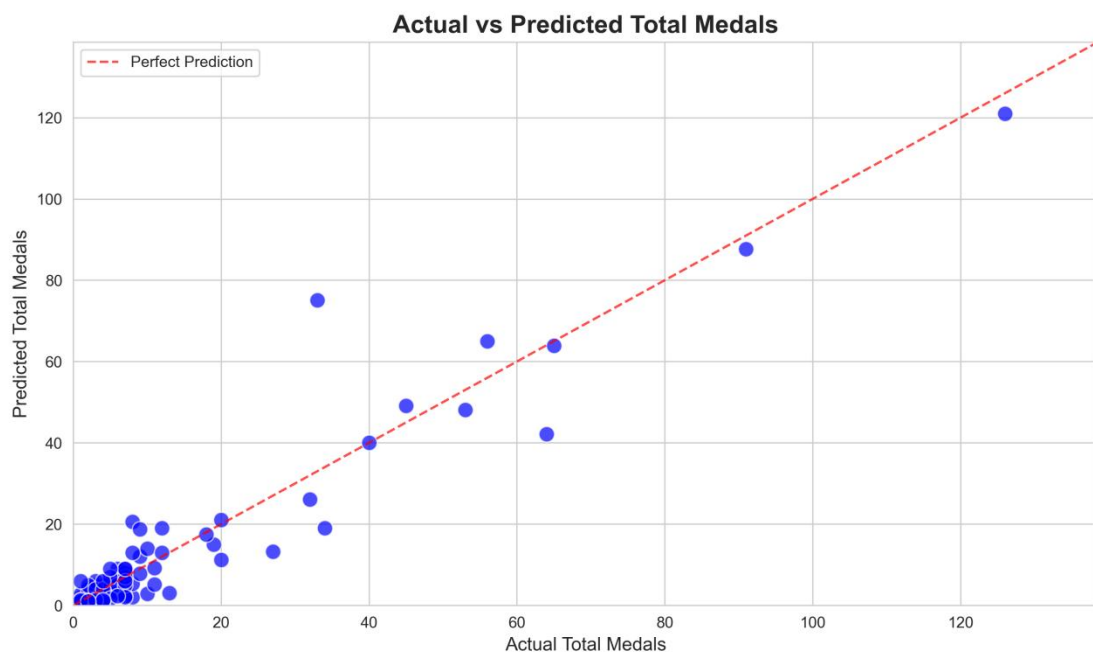


Figure 1: Result of Linear Regression

In **Figure 1**, The red dashed line ($y=x$) indicates perfect prediction alignment. Points closer to the line represent more accurate predictions, while larger deviations indicate higher prediction errors.

Decision Tree

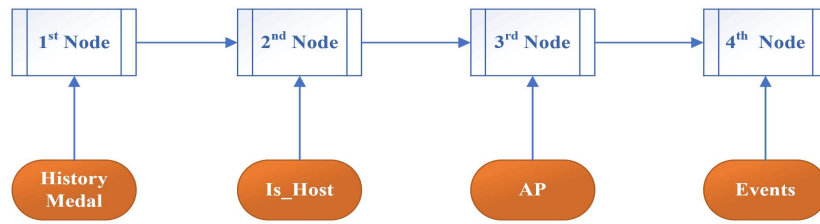
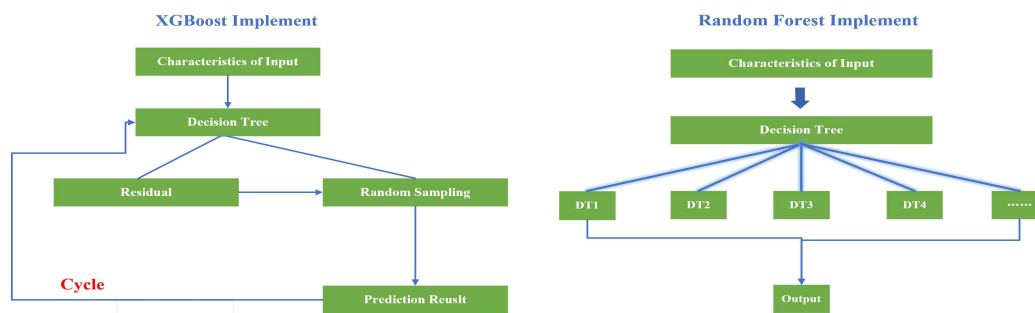


Figure 2:Implement of Decision Tree

As shown in the **Figure 2**, the simple schematic structure of the decision tree contains a total of four variables, and with a single decision tree as the basis, it is possible to construct random forest and XGBoost models(shown in figure 3(a) and figure 3(b)).



(a) the Result of Random Forest on Gold

(b) the Result of XGBoost on Gold

Figure 3:Implement of RF and XGBoost

Through the construction of Random Forest and XGBoost based on decision trees, the model can fit the relationship between variable characteristics and the number of medals, as shown in **Figure 3**.

Table 1:Degree of fit of different models

	Linear Regression	XGBoost	Random Forest
R^2	0.75	0.85	0.84

As can be seen from the **Table 1**, all three models fit well and can respond well to the relationship between the variables and the number of medals, with the linear regression being slightly weaker than the other two models

After each model has predicted the number of medals for each country for 28 years, the predictions of each model for each country are compared and the model with the largest R^2 is selected as the final prediction.

MAPE of the predicted number of total medals for different countries can eventually be obtained.

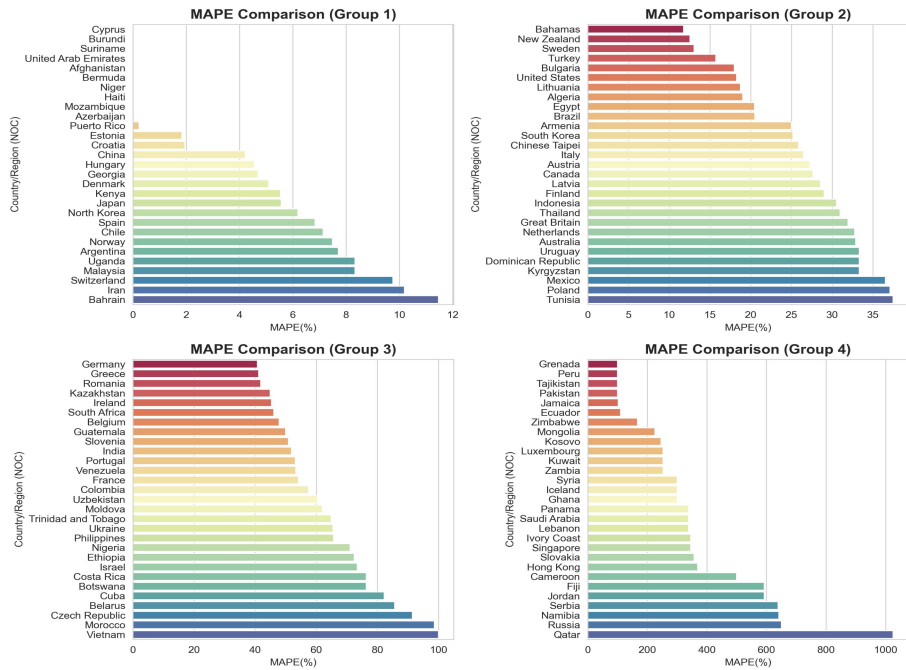


Figure 4: MAPE of the Predict Result

Figure 4 shows the MAPE of different countries grouped into four categories based on the model's prediction performance.

Group 1: Countries with low MAPE, where the model performs accurately, with errors mostly within 10.

Group 2: Countries with slightly higher MAPE, with most errors ranging between 10.

Group 3: Countries with further increased MAPE, where some predictions show errors approaching 100.

Group 4: Countries with the highest MAPE, where errors exceed 500.

4. The Result of Prediction

By comparing between different models, we can get the final accurate prediction results:

Table 2:Prediction of Medal Counts (Top 10)

Country	Gold	Silver	Bronze	Total
United States	43	35	35	113
China	33	27	26	86
Great Britain	26	21	20	67
Japan	16	14	15	45
Germany	13	12	13	38
Australia	12	11	12	35
France	11	11	12	34
Italy	10	10	11	31
South Korea	9	8	9	26
Netherlands	9	8	8	25

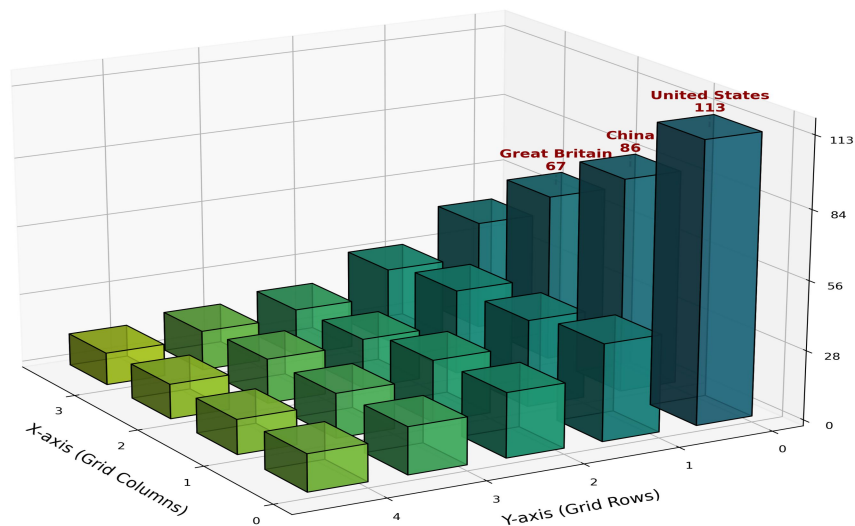


Figure 5:3D Bar Plot of Olympic Medal Counts by Country

Figure 5 displays the total medal counts of different countries, with the X-axis representing grid columns and the Y-axis representing grid rows. The height of each bar corresponds to the total number of medals earned by a country.

United States leads with the highest number of medals (113), followed by China with 67 and Great Britain with 56. The plot allows for a visual comparison of the medal distribution across different countries.

It is required to give intervals for the expected number of medals for each country, which can be restricted to the prediction intervals by the exact predicted values with the residuals.

The formula for the prediction range is given by:

$$\hat{A}_i \pm 1.96 \cdot \sigma_{\text{resid}} \quad (6)$$

Where:

$\hat{A}_i$: This is the predicted value for the i-th observation based on the model.

1.96: This is a constant, corresponding to the critical value for a 95% confidence level from the standard normal distribution.

σ_{resid} : This represents the residual standard deviation, which is a measure of the variability or uncertainty in the model's predictions.

The formula can be interpreted as saying that the true value for the i-th observation is expected to lie within a range around the predicted value, with an upper and lower bound defined by this equation. The factor 1.96 gives a 95% confidence interval, assuming that the residuals (errors between the predicted and actual values) are normally distributed.

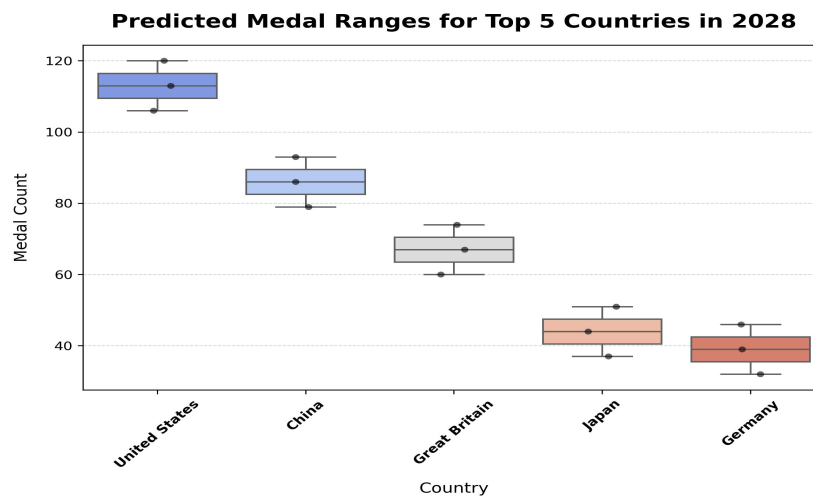


Figure 6: Predicted Range of Total Medals

Figure 6 illustrates the predicted total medal count ranges for the top five countries in the 2028 Olympics.

The United States is predicted to earn the most medals, with the highest median value and the broadest range, indicating high confidence in its performance. China and Great Britain follow with lower, but still substantial, predicted medal counts, each with moderate variability. Japan and Germany show narrower ranges, suggesting more consistency in their expected medal counts, but with fewer total medals compared to the other top nations. The box plot provides a visual representation of the uncertainty in the medal predictions for each country, showing both the expected value (median) and the range of possible outcomes (from the lower quartile to the upper quartile, plus the potential outliers).

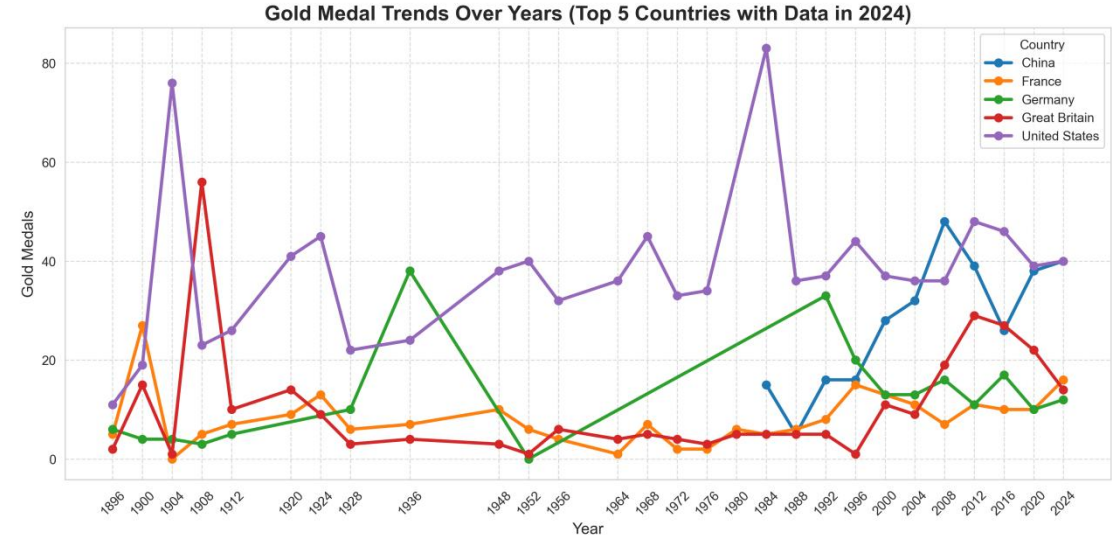


Figure 7: Gold Medal Trends Over Years (Top 5 Countries with Data in 2024)

Figure 7 shows the gold medal trends for the United States, China, Great Britain, France, and Germany from 1900 to 2024. It highlights the fluctuations in each country's Olympic performance over the years.

United States: The U.S. has consistently led in gold medals, showing steady growth, with noticeable peaks in the mid-20th century and recent decades.

China: China experienced rapid growth in gold medals, especially from the 2000s onwards, reflecting a strong focus on sports development.

Great Britain: Great Britain has seen steady progress, with a notable rise since the 2000s, particularly after hosting the 2012 London Olympics.

Germany: Germany's performance has fluctuated, with a decline after reunification but remaining competitive with strong performances in specific sports.

France: France shows stable gold medal counts, but without dramatic increases, maintaining a consistent position in the middle tier.

Total Evaluation:

China stands out for its remarkable rise in gold medals, especially post-1990s. The United States maintains its lead with consistent growth. Great Britain has shown significant improvement, particularly in recent years. Germany has fluctuated but remains competitive. France shows moderate progress without the same rapid growth as China or Great Britain.